



Project Report

Fruit Quality Grading System

Development Track Report

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1. Problem Statement

In Pakistan, the agricultural and food supply industry faces many issues in maintaining a consistent fruit grading and classification system. Fruit sorting is done manually by people, which makes the whole process sluggish and increases the probability of error, especially when different fruit types have almost same physical attributes. Automated classification of these fruits can reduce the involvement of humans, increase standards of these grading and can work as an assistant for exporters, retailers, and quality control systems.

To solve this problem our project focuses on developing a machine-learning-based fruit quality grading system. The main objective is to predict the fruit type using different quality related features such as size, weight, ripeness, sweetness, and defect measurements. Along with predictions, this project also provides useful insights through better visualizations and understanding on Power BI dashboards.

2. Dataset Description

The dataset which has been used in the Fruit Quality Grading System project is fruits.csv, which nearly contain around **10,000 records**. Each record contains and represents one fruit sample and includes multiple qualities related to fruit features.

2.1 Dataset Structure

- **Rows:** 10,000
- **Columns:** 14

2.2 Features

The dataset includes categorical as well as numeric features:

Categorical Features

- fruit_id (identifier)
- fruit_type (targeted variable)
- grade (quality class)

Numerical Features

- size_cm
- weight_g
- color_intensity
- ripeness_level
- firmness
- sweetness
- defect_count
- bruise_score

- spot_score
- defect_score
- size_to_weight_ratio

2.3 Target Variable

The target variable the model is used to predict is the **fruit_type**. It is a multi-class classification problem with nearly 10 different fruit classes. The fruit categories contained in the dataset are Apple, Banana, Watermelon, Pomegranate, Peach, Pear, Orange, Mango, Strawberry, Grapes.

For machine learning model training, the target variable **fruit_type** was first changes into numeric values using **Label Encoding by assigning**. Each fruit type was assigned to a class label ranging from **0 to 9** respectively.

2.4 Domain Context

This dataset is very helpful for real-world agricultural and food supply applications such as:

- food supply chain monitoring
- quality control pipelines
- export grading systems
- automated sorting systems

3. Methodology

Our Fruit Quality Grading System follows a standard machine learning pipeline. This complete pipeline includes data preprocessing, exploratory data analysis (EDA), feature engineering, model training, tuning, evaluation, and dashboard development.

3.1 Data Preprocessing & Understanding (7.1)

Firstly, we explored the dataset to have a grip on its structure and quality. We inspected the data types, missing values, duplicates and class distribution.

3.1.1 Data Understanding:

The dataset was checked for:

- numeric and categorical features
- missing values
- duplicates records
- distribution of target classes

The results displayed that the dataset was clean and balanced, with no missing values or any duplicates.

3.1.2 Missing Value Handling:

After inspection the dataset did not contain missing, we still accumulate missing value handling steps to make the pipeline more reliable:

- Numerical features were filled using **median**
- Categorical features were filled with **mode**

The reason behind is the median that handles outliers better while the mode maintains the original category distribution.

3.1.3 Outlier Detection and Treatment

Outliers are detected using:

- Boxplots
- IQR (Interquartile Range) methods

Naturally fruits vary a lot (for example; watermelon vs grapes are not comparable directly), removing global outliers could exclude valid samples. So, for that reason, we applied to group-wise IQR capping within each fruit type instead of cascading outliers.

3.1.4 Data Transformation

Below are transformations that were applied:

- Label Encoding for fruit_type
- Log transformation ($\log(1+x)$) for highly skewed numerical features
- Feature scaling using robust scaling or min-max scaling, depending on model needs

3.1.5 Data Splitting

The dataset was split using stratified sampling to maintain equal class distribution:

- 70% Training
- 15% Validation
- 15% Testing

This makes sure that the evaluation remains unbiased.

3.2 Exploratory Data Analysis (7.2)

EDA was performed to verify feature distributions along with relationships using:

- histograms
- scatter plots
- boxplots

- correlation analysis

The findings from EDA are:

- weight and size were highly influential in fruit classification
- some fruit classes overlap heavily in size and weight, making classification difficult
- defect-related features were less important compared to physical features

3.3 Feature Engineering & Selection (7.3)

3.3.1 Feature Engineering

These three new features were made:

1. Sweetness–Firmness Ratio

$\text{sweetness_firmness_ratio} = \text{sweetness} / \text{firmness}$

This function helps to capture the relationship between sweetness and firmness, which is linked to ripeness.

2. Defect Density

$\text{defect_density} = \text{defect_count} / \text{size_cm}$

This helps us to normalize the defect count based on the size of fruit.

3. Ripeness-Sweetness Interaction

$\text{ripeness_sweetness_score} = \text{ripeness_level} \times \text{sweetness}$

This displays fruit maturity more effectively.

This function helped the model adapt hidden relationships in the dataset.

3.3.2 Feature Selection

Analysis of the importance of features by using Random Forest showed that the top features are:

- weight_g
- size_cm
- size_to_weight_ratio

These three features contributed around 95% of total predictive importance.

3.3.3 PCA (Dimensionality Reduction)

PCA was applied with 95% variance retention to lower the repetition and multi-collinearity. We compared:

- baseline model
- selected feature model
- PCA-reduced model

PCA reduced complexity while keeping accuracy acceptable.

4. Results (Modeling + Evaluation)

4.1 Model Implementation (7.4.1)

We trained 3 machine learning models:

Model	Category
Logistic Regression	Linear model
Decision Tree	Non-linear tree model
Random Forest	Ensemble learning model

4.2 Hyper-parameter Tuning (7.4.2)

Grid Search was used on Random Forest to improve results.

Best Parameters

- max_depth = 10
- min_samples_split = 5
- min_samples_leaf = 1
- n_estimators = 200

Best CV score achieved: **0.742**

This tuning helped in improving generalization by taking control of tree complexity.

4.3 Cross Validation (7.4.3)

5-fold cross validation was performed:

Logistic Regression Mean Accuracy: 0.7389

Tuned Random Forest Mean Accuracy: 0.7420

Cross validation confirms that the stable performance is with low variation.

4.4 Model Interpretation (7.4.4)

- **Random Forest Feature Importance**

Top three features:

Feature	Importance
weight_g	40.28%
size_cm	29.39%
size_to_weight_ratio	25.42%

This c shows classification majorly depends on the physical measurements.

- **Logistic Regression Coefficients**

Logistic regression coefficients also support the exact same observation, showing that size, weight, and ratio values dominate the prediction process.

5. Model Evaluation (7.5)

5.1 Overall Performance

All models were able to achieve accuracy close to 74%.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	74.00%	73.64%	74.00%	72.66%
Decision Tree	73.93%	74.01%	73.93%	73.78%
Random Forest	74.13%	73.72%	74.13%	73.57%

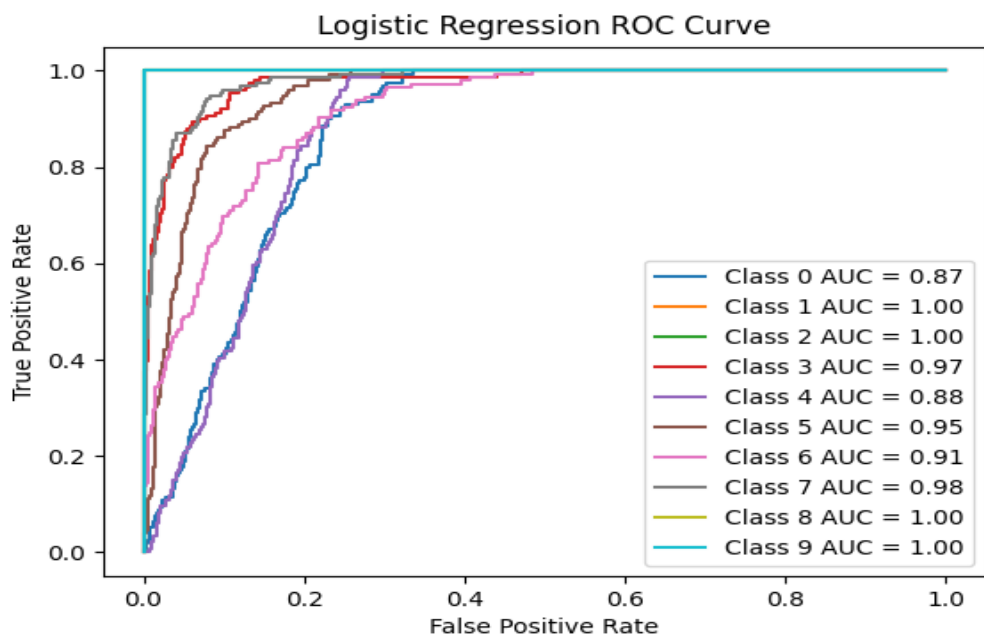
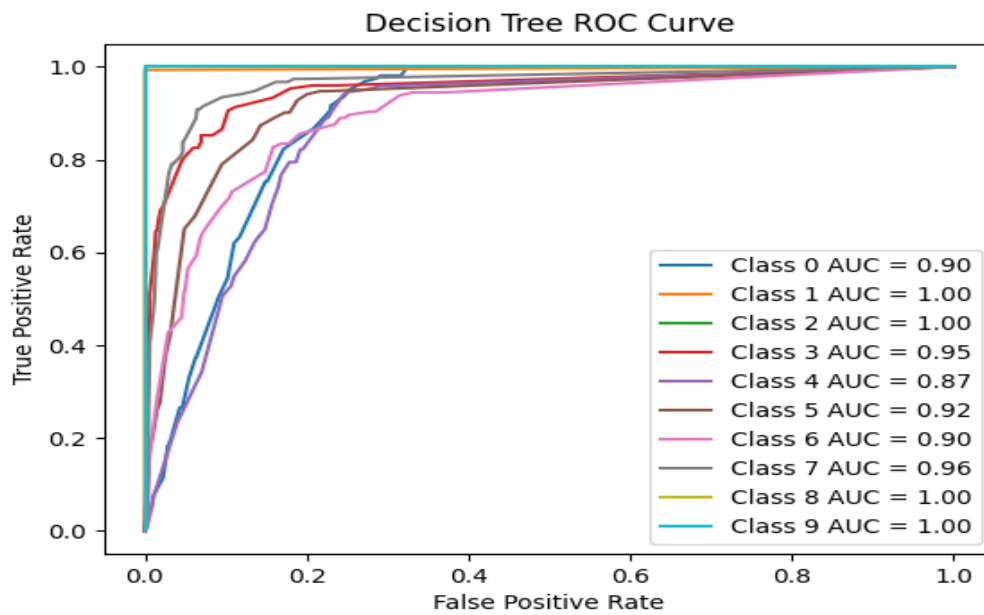
Random Forest achieved a significant overall performance.

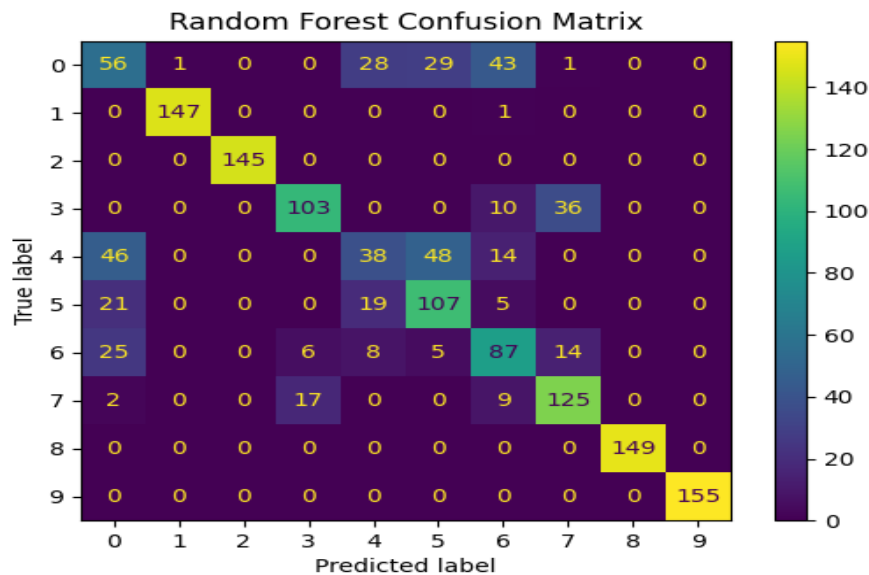
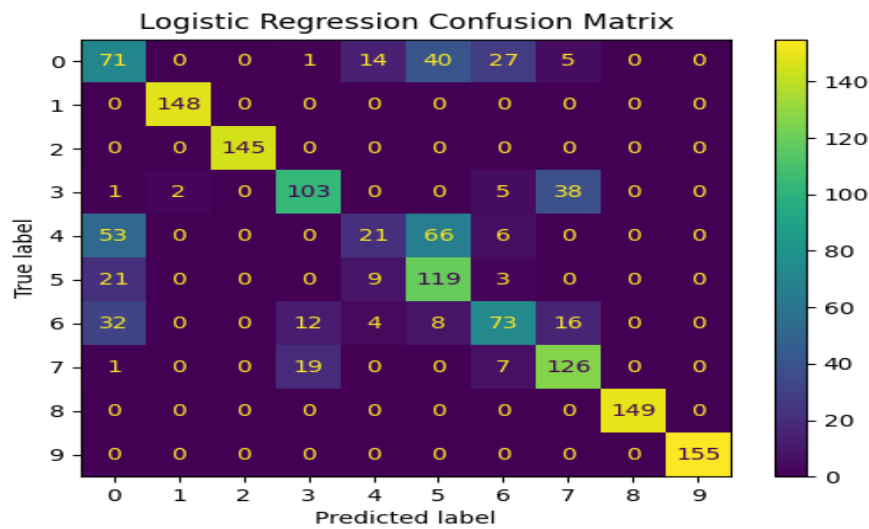
5.2 ROC Curve and AUC

AUC results present good class separability.

- Classes **1, 2, 8, 9** had AUC = **1.000**
- Difficult classes were **0 and 4** with AUC around **0.87–0.89**

Random Forest displayed the most consistent AUC performance across classes.





5.4 Final Model Selection

The last selected model is **Tuned Random Forest Classifier** because it delivers:

- handles non-linear relationships better than Logistic Regression
- best cross validation performance
- strongest ROC/AUC stability
- highest test accuracy

6. Dashboard Explanation (Power BI)

The reason to create Power BI dashboard was to visualize dataset insights and model overall performance. The dashboard consists of three pages

Page 1: Overview Dashboard

Focus: dataset summary

Includes:

- average weight and size per fruit type
- slicers for fruit type and grade
- fruit type distribution chart
- grade distribution chart
- KPI cards (total samples, fruit classes, average weight, average size)

Purpose: provides an overall view of what the dataset contains.



Page 2: EDA Dashboard

Focus: feature behavior and relationships

Includes:

- histograms of size along with weight
- scatter plot:
 - size vs weight

- sweetness vs firmness
- sweetness vs ripeness
- sweetness vs color intensity

Purpose: describes how different features behave and which ones affect classification the most.



Page 3: Model Insights Dashboard

Focus: model comparison and interpretability

Components:

- KPI cards
- model comparison chart (accuracy, precision, recall, F1)
- feature importance chart
- F1 score per class per model chart
- importance bar chart for features
- summary panel

Purpose: displays which model performed best and the reason behind it.

